DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2019/2020

SECOND YEAR SEMESTER II EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE

CCS 2208: COMPUTER GRAPHICS

DATE: AUGUST 2019 **TIME: 2 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question One: (30 Marks)

a) Describe three advantages of using Graphical User Interfaces. **(3 marks)**

b) Describe three uses of chromaticity diagram **(3 marks)**

c) Distinguish between the following terms
   i. Emissive and non-emissive displays **(2 marks)**
   ii. Simulation and animation **(2 marks)**
   iii. Window port and view port **(2 marks)**

d) The Phongs reflection model is a sum of three terms that each models a specific type of light reflection. Name and describe the three terms. **(3 marks)**

e) Explain the following OpenGL functions. **(4 marks)**
   i. glClear(GL_COLOR_BUFFER_BIT);
   ii. glVertex2i(12,30);
   iii. glClearColor (1.0,1.0,0.7,0.1);
   iv. glDisplayMode(object);

f) Mary wants to acquire a graphics display device, explain four factors she needs to consider in the acquisition process. **(4 marks)**

g) Given a triangle with vertices (2,2), (4,2) and (4,4), find the transformed coordinates after a 90° rotation about the origin. **(3 marks)**

h) Suppose you have an image which you want to store in permanent storage, describe four factors you would consider when selecting a file format. **(4 marks)**

Question Two: (15 Marks)

a) Define affine transformation giving at least one property. **(2 marks)**

b) Given a polygon with the vertices (0,0), (0,1), (1,0) and (1,1). Draw the shape obtained after transforming this unit square with the homogeneous matrix **M** given below. **(3 marks)**

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(90) & -\sin(90) & 0 \\ \sin(90) & \cos(90) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

c) Rasterize a line having end points (10,20) and (16,25) and plot the resultant pixels using Bresenham's algorithm. **(4 marks)**

d) Suppose you are given a scene to render through the graphics pipeline. Explain the Cartesian reference frames and procedure you would use to construct and display the scene. **(6 marks)**

Question Three: (15 Marks)

a) Briefly explain the purpose of the following methods used in creating 3-D sceneries
   i. Parallel projection **(1 mark)**
   ii. Perspective projection **(1 mark)**

b) Explain two types of texture mapping. **(2 marks)**

c) Given 3D geometric primitives explain how you would generate an image using the 3D rendering pipeline. **(6 marks)**

d) Write a simple OpenGL program for rendering on a black background a red square. **(5 marks)**

Question Four: (15 Marks)

a) Consider two raster systems with the resolutions of 1280 x 720 and 2048 x 1536.
   i. Describe two ways of refreshing a display screen. **(2 marks)**
   ii. How many pixels could be accessed per second in each of these systems by a display controller with a refresh rate of 120Hz? **(2 marks)**
   iii. What is the access time per pixel in each system? **(2 marks)**

b) Film animators use a variety of methods for depicting and emphasizing motion sequences. Discuss three traditional animation techniques giving examples. **(6 marks)**

c) How is colour represented in computer graphics and how does this relate to the Human Visual System. **(3 marks)**